

HOMEWORK 7

1. Let K/F be a field extension, $\alpha, \beta \in K$. Show that if $[F(\alpha) : F]$ and $[F(\beta) : F]$ are relatively prime, then $[F(\alpha, \beta) : F] = [F(\alpha) : F] \cdot [F(\beta) : F]$.
2. Find the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$.
3. Find the minimal polynomial over $\mathbb{Q}$ of a (complex) primitive root of unity of degree p^n (p is prime).
4. Let x be a variable over a field F and $y = \frac{x^2}{x-1}$. Find $[F(x) : F(y)]$.
5. Let K be the subfield of $\mathbb{C}$ generated by all complex roots of the polynomial $X^3 - 2$ over $\mathbb{Q}$. Find $[K : \mathbb{Q}]$.
6. Let K/F be a field extension and $\alpha, \beta \in K$ two elements that are algebraic over F . Prove that if α and β have the same minimal polynomials over F then the fields $F(\alpha)$ and $F(\beta)$ are isomorphic over F .
7. Find an example of a nontrivial field extension K/F with isomorphic fields K and F .
8. Let K/F be a field extension, $\alpha \in K$. Show that if $f(\alpha)$ is algebraic over F for some non-constant polynomial $f \in F[X]$, then α is also algebraic over F .
9. Let K/F be a field extension, $\alpha, \beta \in K$. Prove that $[F(\alpha, \beta) : F(\alpha + \beta, \alpha\beta)] \leq 2$.
10. Find example of an infinite algebraic extension.